GIS APPLICATIONS IN TELECOMMUNICATIONS

5-Day Intensive Training Program

DAY 1 LAB MANUAL

GIS Fundamentals, Data Types & Geodatabase / ER Cardinality Design

Version 3.0 | Instructor-led Hands-on Manual + Companion Presentation

Platform: ArcGIS Pro (ArcMap concepts referenced)Duration: 1 Day (09:00 – 16:00)Scenario: MetroTel National ISP — From Core POP to Homes & Towers

Companion deck: Day1_GIS_Telecom_Concepts_Presentation_v3.pptx

Document	Detail
Course	GIS Applications in Telecommunications
Day	Day 1 — GIS Concepts, Components, Data Fundamentals & Geodatabase Design
Version	3.0
Primary tools	ArcGIS Pro Catalog/Map, Create Feature Class/Dataset, XY Table To Point, Define Projection/Project, Relationship concepts / cardinality design
Presentation	Day1_GIS_Telecom_Concepts_Presentation_v3.pptx
Data folder	Day1_GIS_Fundamentals/data/

1. Learning Objectives (Hook + Skills)

By the end of Day 1, participants will be able to:

- Explain what GIS is and why telecom planning collapses without location.
- Define core GIS building blocks: features, attributes, layers, CRS, raster vs vector.
- Distinguish geographic vs projected coordinate systems and choose when to project.
- Design a file geodatabase with feature datasets and feature classes.
- Define entities and relationships for an ISP network and assign correct cardinality (1:1, 1:M, M:N).
- Map the real device chain: National ISP Core → Hub → OLT → FDH → Splitter → Home, plus Tower → Sectors.
- Load synthetic MetroTel data and prove relationships with selections/joins.

2. Day Agenda

Time	Session	Outcome
09:00–09:40	Hook + Why GIS for telecom	Motivation & vocabulary start
09:40–10:45	Lab 1 — GIS anatomy in ArcGIS Pro	Map, layers, attributes, symbology

10:45–11:00	Break	—
11:00–12:15	Lab 2 — CRS, vector/raster, XY points	Correct CRS + towers/homes on map
12:15–13:15	Lunch	—
13:15–14:45	Lab 3 — Geodatabase + ER cardinality design	GDB design + ISP hierarchy model
14:45–15:00	Break	—
15:00–15:45	Lab 4 — Prove relationships in data	1:M selections, join tests
15:45–16:00	Recap quiz + preview Days 2–5	Day 1 locked in

3. Concept Definitions (Study Notes — Learn These Cold)

3.1 GIS core

Geographic Information System (GIS): A system to capture, store, query, analyze, and visualize data that has a location on Earth — both WHERE something is and WHAT it is.

Spatial data: Data with geometry (point/line/polygon/raster) referenced to a coordinate system.

Attribute data: Tabular descriptors of a feature (e.g., TOWER_ID, HEIGHT_M, STATUS).

Feature: One real-world object represented as geometry + attributes (one tower, one cable, one district).

Layer: A display/analysis unit of features of the same type (Towers layer, Fiber layer).

Basemap: Reference background (streets/imagery) for context; not usually your analysis layer.

3.2 Vector vs raster

Vector: Discrete geometries: points (towers/homes), lines (fiber), polygons (districts). Best for networks and assets.

Raster: Grid of cells (pixels), each with a value (elevation, coverage strength). Best for surfaces/continuous fields.

Topology: Rules about how features share geometry (connect, contain, not overlap) — critical for networks.

3.3 Coordinate systems

Geographic Coordinate System (GCS): Location as latitude/longitude on a spheroid/datum (e.g., WGS84). Units = degrees. Good for storage/GPS exchange.

Projected Coordinate System (PCS): Flattened map coordinates in linear units (meters/feet), e.g., UTM. Required for accurate distance, buffer, area.

Datum: Reference model of Earth used by a CRS (e.g., WGS84). Mixing datums without transform causes shifts.

Define Projection vs Project: Define Projection only labels data that already has correct coordinates but missing metadata. Project mathematically converts coordinates to another CRS.

3.4 Geodatabase building blocks

Geodatabase (GDB): Esri's spatial database container for feature classes, tables, relationship classes, domains, and topologies.

Feature Dataset: A folder-like container inside a GDB that shares one spatial reference; used to group related feature classes (e.g., FD_ISP_Access).

Feature Class: A homogeneous collection of features with the same geometry type and schema (Towers points, Fiber lines).

Object Class / Table: Non-spatial table in the GDB (e.g., ServicePlans).

Domain: Allowed values/ranges for a field (e.g., STATUS = Active/Planned). Improves data quality.

Subtype: Categories within a feature class with different defaults/domains (e.g., TowerType Rooftop vs Greenfield).

3.5 Database / ER / cardinality (critical for afternoon)

Entity: A thing we store (Tower, Splitter, Home). In GIS often a feature class or table.

Relationship: A business rule linking entities (Splitter serves Homes).

Primary Key (PK): Unique ID of an entity row (TOWER_ID, HOME_ID).

Foreign Key (FK): Field storing another entity's PK to create a link (Homes.SPLITTER_ID → Splitter.NODE_ID).

ER Diagram: Entity–Relationship diagram: boxes = entities, lines = relationships, symbols = cardinality.

Cardinality: How many instances of entity A relate to how many of entity B.

1:1 (one-to-one): Each A relates to at most one B, and each B to at most one A. Example (simplified): Customer ↔ Primary Service Account.

1:M (one-to-many): One A relates to many B; each B relates to one A. Example: One Splitter → many Homes; One Tower → many Sectors.

M:N (many-to-many): Many A relate to many B. Requires a bridge table. Example: Towers ↔ Landlords over multiple lease periods.

Relationship Class: Geodatabase object that stores/enforces relationships between classes (can be origin-destination with cardinality).

NOTE: In telecom GIS, most physical hierarchy is 1:M. If you model M:N without a bridge table, queries and inventory will break.

3.6 Telecom device chain used in Day 1

Fiber / fixed access chain (parent → child, all 1:M):

- National ISP Core POP → Regional Hub → OLT/Exchange → FDH/Cabinet → Splitter → Home/ONT

Mobile / wireless chain:

- Backhaul Node (Hub/OLT) → Tower → Sector (and technology carriers)

This is the story on the cardinality ER slide in the presentation. Memorize it.

4. Datasets (Synthetic + Real options)

File	Type	Purpose
ISP_NetworkNodes.csv	Points hierarchy	Core/Hub/OLT/FDH/Splitter nodes
FiberCables.geojson	Lines	Cables linking parent-child nodes
Homes_ONT.csv	Points	Homes/ONTs with SPLITTER_ID (1:M child)
Towers.csv	Points	Towers with BACKHAUL_NODE
Sectors.csv	Table/points optional	Sectors with TOWER_ID (1:M)
Districts.geojson	Polygons	Simple admin areas for spatial context
Cardinality_Rules_ER.csv	Rules table	Official cardinality definitions for design
CRS_Examples.csv	CRS cheat sheet	When to use WGS84 vs UTM
Day1_AllTables.xlsx	Excel	Instructor workbook

REAL_DATA_SOURCES.txt	Text	Optional real open data

5. Lab 1 — GIS Anatomy Hook (ArcGIS Pro)

Objective: See telecom assets as map + table at the same time — the GIS “aha”.

- New project MetroTel_Day1_Fundamentals.aprx; create MetroTel_Day1.gdb
- Add Districts.geojson; symbolize uniquely by DIST_NAME; transparency 40%
- XY Table To Point: ISP_NetworkNodes.csv (LON/LAT, GCS WGS84) → ISP_Nodes_WGS84
- Unique value symbology on NODE_TYPE (Core, Hub, OLT, FDH, Splitter)
- Add FiberCables.geojson; categorize by CABLE_TYPE (Backbone/Aggregation/Feeder/Distribution)
- Open attribute table of ISP_Nodes; Select NODE_LEVEL = 1; flash on map
- Select NODE_TYPE = 'Splitter' and discuss: these are the last shared optical points before homes

CHECKPOINT: Students can click a cable/node and explain what real device it is.

6. Lab 2 — CRS, Vector Types, XY Homes & Towers

- XY Table To Point: Towers.csv → Towers_WGS84; Homes_ONT.csv → Homes_WGS84
- Add Sectors.csv as table (non-spatial for now)
- Measure a rough distance between two towers in WGS84 (degrees) — show why degrees are bad for meters
- Project tools: Project Towers/Homes/Nodes/Cables/Districts to WGS84 UTM Zone 42N (or local UTM) into FD later
- Buffer a tower by 500 m in UTM; show valid meter buffer
- Identify geometry types present: points (nodes/towers/homes), lines (fiber), polygons (districts)

TIP: Rule of thumb: store exchange in WGS84 if needed, analyze in projected meters.

CHECKPOINT: Homes + towers visible; student can state GCS vs PCS difference in one sentence.

7. Lab 3 — Geodatabase Design + ER Cardinality Workshop

Objective: Design the GDB like a telecom inventory brain, not a folder of shapefiles.

- Create Feature Dataset FD_ISP_Network with UTM spatial reference
- Load projected feature classes: ISP_Nodes, FiberCables, Homes, Towers, Districts into FD_ISP_Network
- Import Sectors as table Sector_Inventory into GDB
- On whiteboard/PPT ER slide, draw entities and mark cardinality using Cardinality_Rules_ER.csv
- Implement FK fields already present: Homes.SPLITTER_ID, Sectors.TOWER_ID, Towers.BACKHAUL_NODE, Nodes.PARENT_ID

- Optional: Create Relationship Class RC_Splitter_Homes (Origin=Splitters query/view or nodes subtype, Destination=Homes, cardinality One-to-Many)
- Optional: Create Relationship Class RC_Tower_Sectors (1:M)
- Add Domain Dom_Status: Active, Planned, OnAir; assign to STATUS fields where applicable

NOTE: If Relationship Class UI is time-limited, still complete the ER diagram and FK mapping — that is the core Day 1 competency.

CHECKPOINT: ER diagram completed with 1:1 / 1:M / M:N labels; GDB contains FD + feature classes + sectors table.

8. Lab 4 — Prove Cardinality with Data

- Select one splitter (e.g., SPL-01). Select homes where SPLITTER_ID = that ID. Count children → demonstrates 1:M
- Select one tower; select sectors with that TOWER_ID → 1:M Tower–Sector
- Select ISP-CORE-01; select nodes with PARENT_ID chain step-by-step down to splitters (hierarchical drill-down)
- Join Sectors table to Towers (TOWER_ID) and calculate how many sectors per tower (Summary Statistics)
- Spatial Join Towers to Districts (Intersect) → which district contains which towers (spatial 1:M)
- Discuss M:N example: if two landlords can lease parts/time on many towers, you need LeaseBridge(TOWER_ID, LANDLORD_ID, START, END)

CHECKPOINT: Student can explain why Home does not store a list of splitters, and why Sector stores TOWER_ID.

9. Expected Outputs Checklist

#	Deliverable	Done?
1	MetroTel_Day1_Fundamentals.aprx + MetroTel_Day1.gdb	
2	Map with districts, ISP nodes, fiber, towers, homes	
3	Projected UTM copies used for buffer/measure	
4	Feature Dataset FD_ISP_Network populated	
5	Completed ER diagram with cardinality (photo/PDF)	
6	Demonstrated 1:M selections (Splitter→Homes, Tower→Sectors)	
7	Day 1 quiz completed	

10. Quick Quiz

- Define GIS in one sentence.
- Vector or raster for fiber cables? Why?
- When must you use a projected CRS?
- What cardinality is Splitter → Homes? Where is the foreign key stored?

- What cardinality is Tower → Sector?
- Why is Tower ↔ Landlord often M:N?
- Name the device chain from national ISP core to the home.

Answers:

- 1) System for location + attribute data to map/analyze decisions.
- 2) Vector lines — discrete network assets.
- 3) Distance/area/buffer/network length in meters.
- 4) 1:M; FK on Homes (SPLITTER_ID).
- 5) 1:M; FK on Sectors (TOWER_ID).
- 6) Many towers can relate to many landlords across leases; needs bridge table.
- 7) Core POP → Regional Hub → OLT → FDH → Splitter → Home/ONT.

11. Bridge to Day 2

Day 2 uses these GIS fundamentals to map 2G/3G/4G coverage, find gaps, and support site selection with Spatial Analyst / suitability tools. Clean tower/sector schemas from Day 1 become Day 2 inputs.

12. Version History

Version	Changes
v1	Draft GIS fundamentals outline
v2	Added GDB design labs
v3	Full manual + strong concept definitions, ISP-to-home & tower-sector cardinality model, synthetic datasets, companion presentation with dedicated ER cardinality slide

— End of Day 1 Manual (Version 3.0) —